

**ADMINISTRATIVE INFORMATION****FMI No: 25300****FMI Title: LightSpeed Ultra/Plus Ver. 303 S/W Upgrade**

FMI TYPE	CHARGE CODES	IMPLEMENTATION DATES
☒ Mandatory Safety	GEMS - AM:	Issue Date:
☐ Mandatory	GEMS - E:	
☐ Mandatory on Request	GEMS - A:	Close Date:

- ☐ **Staged FMI** **Customer List** provided by FMI Administration. If you identify additional units within effectivity range, but they are not shown on the customer list, contact FMI Administration in your Pole.
- ☐ **Parts** and any special tools are furnished automatically. To address shortages and ordering exceptions, contact FMI Administration in your Pole.
- ☐ **Parts Ordering:** GEMS - Americas, Contact your Region logistics coordinator
GEMS - Europe, Order FMI kits from GPO/EPL Buc
GEMS - Asia, Contact your region logistics coordinator
- ☐ **No Parts** are required to complete this FMI.
- ☒ **Batch with** FMI 25298, FMI 25299, FMI 25307
- To minimize costs, combine implementation of this FMI with the FMI's listed for all customers where the effectivity is the same (Please see your customer lists).
- ☐ **Non-Staged** **No Customer List is provided.** Each service location must identify affected units, and place part orders as indicated in the Special Instructions.

Technical Support

GEMS - Americas	GEMS - Europe	GEMS - Asia
GE Medical Systems Insite Center (414) 524-5300 800-321-7973 Milwaukee, WI 53201, USA	GEMS - E / ESC BP34 F 78533 BUC CEDEX, FRANCE Tel: (33 1) 39 20 00 07 or ESC Fax: (33 1) 30 70 99 70	Asia Support Center Phone: 81-426-56-0033 Fax: 81-426-48-2905

Special Instructions:

For tracking purposes the Gantry (model 2281177 manufactured before February 15, 2002) will be used as the locating identifier. This FMI applies to all LightSpeed Ultra/Plus Systems with Software model numbers 2305095-2, 2305095-3, 2305095-4.

Unit Tracked: 2281177**FMI Instructions Part No.: 2329509-100****Do NOT leave unsecured on site**

FMI Completion Certification Report

Mail To

GEMS - Americas

G.E. Medical Systems
Attn: FMI Admin. W-523
P.O. Box 414
Milwaukee, WI 53201

GEMS - Europe

GEMS - Europe
FMI Admin. 322
BP34
F78533 Buc Cedex, France

Send to your
Country FMI or Service
Administrator

FMI No.: 25300

FMI Title: LightSpeed Ultra/Plus Ver. 303 S/W

Customer Full Name: _____

Number and Street Address: _____

City and Country: _____

Country Code

Site Number

System I.D.

Unit Tracked:

Model Number	Serial Number	Compl. Date DD/MM/YY	FMI Comp. Code	Service Hours	
				Labor	Travel

FMI Completion Code

1. Modification Complete or previously modified.
2. FMI not required per FMI instruction.
3. Not modified. Unit in storage. FMI kit returned to stock.
4. Modification refused or equipment altered. Customer signature and title required.
5. Unit location unknown or unit not in area. Product Locator notified.

Customer Refusal or Equipment Altered - Modification Pending.

Installer

Comments _____

Effectivity

The effectivity of this FMI includes the following software Model Numbers:

2305095-2, 2305095-3, 2305095-4

To more efficiently track the location of the systems for this FMI, we will use the gantry model number 2281177 with a manufacturing date of February 15, 2002 or earlier.

Purpose

This FMI will bring the installed base LightSpeed Ultra/Plus MDAS systems up to M4 release software status. This release of software is intended to solve quality issue for the installed base products and make a common set of software for the multiple hardware sets currently in the field. The features of this software include (but are not limited to) the following features:

- Quality improvements to the Reconstruction Engine
- Support for the Octane1 Dual Processor Operator Console
- Support for the Octane2 Single Processor Operator Console
- Support for the Octane2 Dual Processor Operator Console
- Support for the 4 Slice (depopulated MDAS)

This upgrade is only intended to operate on LightSpeed Ultra/Plus systems installed with an MDAS detector system. This system can be identified by the gantry model number (2281177).

Prerequisite FMIs

- There are a number of systems that were sold as 4 Slice scanners with a full set of MDAS converter cards. Unless these customers have purchased the 8 Slice option or are running the flex trial FMI 25307 is required. The systems in this group at the time of the printing of this FMI can be identified by comparing the gantry model 2281177 and serial numbers listed below:

272140CN3, 273735CN1, 274017CN3, 274279CN9, 274331CN8, 276194CN8, 276689CN7, 276738CN2, 276851CN3, 276859CN6, 276867CN9, 276920CN6, 276922CN2, 277105CN3, 277150CN9, 277220CN0, 277221CN8, 277759CN7, 277961CN9, 279103CN6, 281421CN8, 282187CN4, 282316CN9, 282336CN8, 282640CN2, 282652CN7, 282653CN5, 282934CN9, 282935CN5, 282937CN2, 283008CN1, 283398CN6, 283398CN6

Related FMIs

The following two FMIs are associated with this FMI due to the needed software changes to support the introduction of hardware in these two FMIs.

- FMI 25298
- FMI 25299

Furnished Materials

FMI 25300 Kit (Part Number 2333320) includes the following items:

Qty	Description	Part #
1	FMI 25300 Document	2329509-100
1	LightSpeed Ultra/Plus Software Load Procedure	2296436-100
1	LightSpeed Ultra/Plus Application Tips & Workarounds	2296447-100
1	LightSpeed Ultra/Plus Serialized Software Set	2305095-5

The following items are included in the Serialized Software Set

Qty	Description	Part #
1	Rating Plate for Serialized Software 2305095-5	
1	LightSpeed Ultra/Plus Serialized Software Set	2305091-5
	Operating System Disk 1	2305087
	Operating System Disk 2	2305088
	Operating System Disk 3	2305089
	Applications Software	2305090-5

Tools Required

Qty	Description	Part #
1	LightSpeed Ultra/Plus Class C	2305092-5
1	LightSpeed Ultra/Plus Class M	2305093-5

People Power

4 hours labor and 1 hour of travel

Compatibility

Opening the Service Desktop and selecting the HOME tab will display system information. In the section labeled "Current System Status" you will find the "Application Software". This FMI applies to systems that have the software version indicated below.

The version displayed for the LightSpeed Ultra/Plus **4 Slice MDAS M3** is:

300.2_2.9C_0628_H34M3

The version displayed for the LightSpeed Ultra/Plus **8 Slice MDAS M3 Octane1** is:

301.1_2.9C_20010628_H3M3

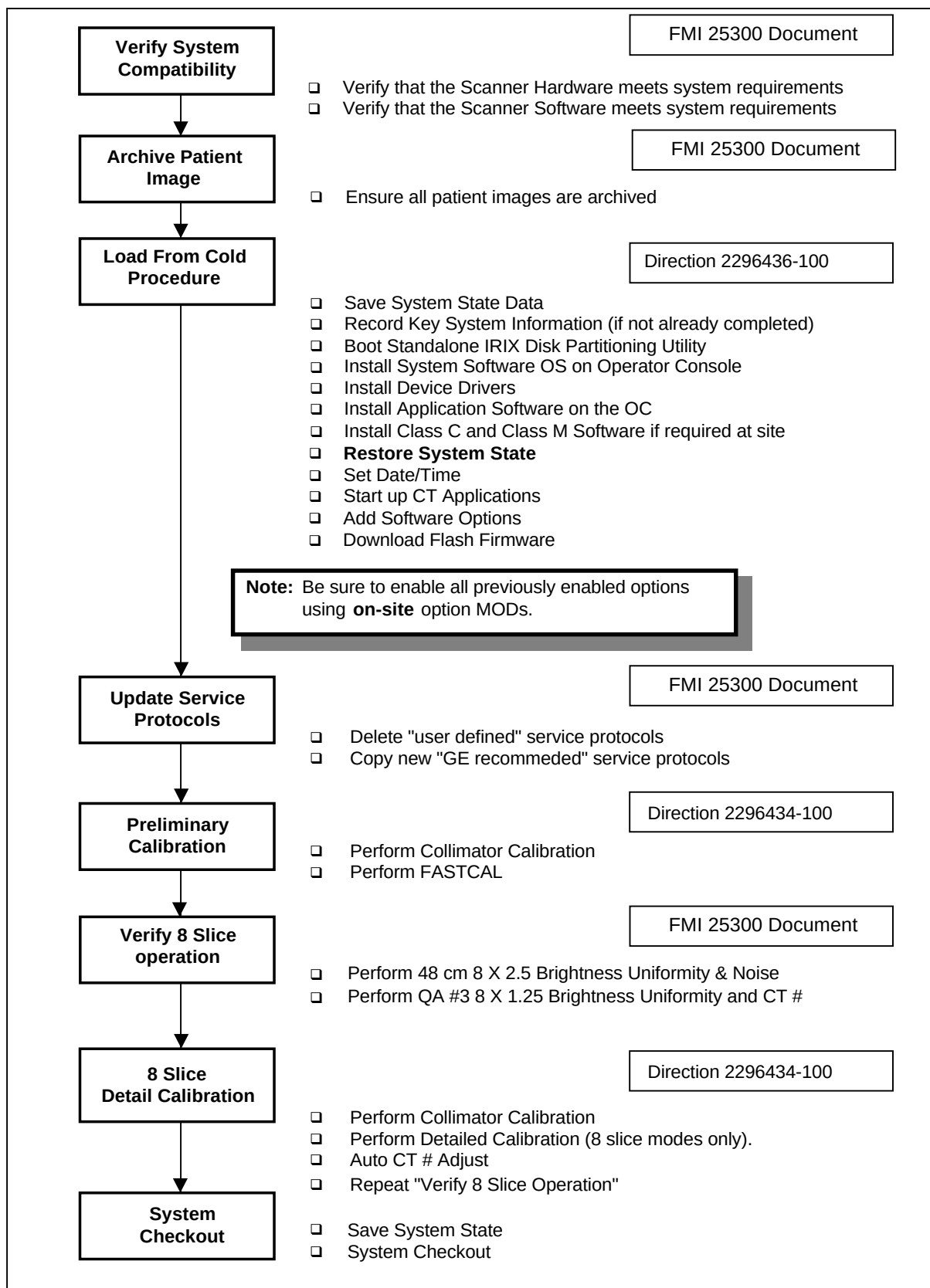
The version displayed for the LightSpeed Ultra/Plus **8 Slice MDAS M3 Octane2** is:

302.1_H3M3.1

Once this FMI is performed both 4 and 8 Slice MDAS is supported and the version will be:

303.1_H3M4

Steering Guide



Procedure

Section 1 - Verify System Compatibility

1. If your gantry model number is **not** 2281177, **STOP** this FMI does not apply.
2. If your gantry model number is 2281177, continue to the next step.
3. Verify that your system is equipped with the 8 Slice Option MOD. If it is not, compare your gantry serial number to that listed in the prerequisite FMI section to see if you need to perform FMI 25307 before continuing with this FMI.
4. Compare your system software to the system software requirements listed in compatibility section.

Open the Service Desktop and select the HOME tab to display system information. In the section labeled "**Current System Status**" you will find the "**Application Software**".

The version displayed for the LightSpeed Ultra/Plus **8 Slice MDAS M3** is:

300.2_2.9C_0628_H34M3

The version displayed for the LightSpeed Ultra/Plus **8 Slice MDAS M3 Octane1** is:

301.1_2.9C_20010628_H3M3

The version displayed for the LightSpeed Ultra/Plus **8 Slice MDAS M3 Octane2** is:

302.1_H3M3.1

5. If the system does not have version 303 or higher software, verify that you have software collector 2305091-5 and the customer specific Options MOD, available for software load.

Section 2 - Archive Patient Images

Verify that the customer has archived all patient data.

Section 3 - Software Load From Cold

Perform the Load From Cold procedure as detailed in Chapter 1 of "LightSpeed Ultra/Plus Software Installation Procedures" (2296436-100).

NOTICE: It is required that a "Restore STATE" be performed following the load of Class C and Class M software. To facilitate this, it is recommended that the Class C and Class M be loaded following the Applications Load (Section 3.6).

NOTICE: A Flash Download must be performed after the 8 Slice option is loaded. In some cases, the subsequent reboot indicates a need to reflash the system. If this occurs you will need to cycle the power to the gantry to clear the problem.

Section 4 - Update Service Protocols

This section details the process for updating protocols that were written to the system as part of the Restore STATE process. A new set of Service Protocols has been created for the support of 8 Slice testing. These have been added to the system as a part of the LFC process. However, they may have been written over by the Restore STATE process. A new feature allows us to selectively copy "GE recommended" protocols into the displayed "User Defined" protocols.

1. With System Applications running on the scanner, select the **Copy Protocols** button from the top right corner of the right monitor. This will open a GUI that is used for updating the protocols.
2. Cursor select the patient image (adult). This will change the patient image to a baby and display the protocols associated with this image.
3. Expand the list under the selection "Infant Upper Extremity" on the User Defined side of the GUI by clicking on the words for this selection. This will display all of the service protocols associated with "Infant Upper Extremity".
4. Delete each of the protocols listed under the User Defined/Infant Upper Extremity directory by selecting each protocol and pressing the **Delete** button.

NOTICE: To copy a protocol from the GE Recommended to the User Defined side of the GUI you must first select the User Defined directory that you wish to copy the protocol into. Then select the GE Recommended protocol to be copied.

5. Copy each of the protocols found in the GE Recommended/Infant Upper Extremity directory to the User Defined/Infant Upper Extremity directory by selecting each protocol and pressing the **Copy** button.
6. Expand the list under the selection User Defined "Infant Miscellaneous" by clicking on the words. This will display all of the service protocols associated with "Infant Miscellaneous".
7. Delete each of the protocols listed under the User Defined/Infant Miscellaneous directory by selecting each protocol and pressing the **Delete** button.

NOTICE: To copy a protocol from the GE Recommended to the User Defined side of the GUI you must first select the User Defined directory that you wish to copy the protocol into. Then select the GE Recommended protocol to be copied.

8. Copy each of the protocols found in the GE Recommended/Infant Miscellaneous directory to the User Defined/Infant Miscellaneous directory by selecting each protocol and pressing the **Copy** button.

Section 5 - Calibration

1. Perform Collimator Calibration by selecting Scanner Utilities on bottom of the left monitor then selecting Collimator Calibration from the displayed selections.
2. Perform FASTCAL by selecting Daily Prep on the bottom of the left monitor, then selecting FASTCAL from the displayed selections.

NOTICE: You may get a pop-up indicating: "System detects an alignment or change was performed on the System". If this happens, press the Ok button and continue with the FASTCAL.

Section 6 - Verify 8 slice Operation

If your system does not have the 8 slice option skip to Section 8.

The functionality of your 8 slice option can be tested by performing the detailed testing outlined in Appendix A of this document. This process uses the 48cm Body Phantom and the 20cm QA phantom. Two acquisitions are performed and then the images are processed and evaluated per the instructions in Appendix A.

If the evaluation process meets the specifications detailed in Appendix A, skip Section 7 and go directly to Section 8.

If the specifications are not met in the testing performed in Appendix A, proceed to Section 7 and perform a Detailed Calibration.

Section 7 - Detailed Calibration of 8 slice Modes

1. Perform DAS Gain Calibration by selecting Scanner Utilities on bottom of the left monitor then selecting DAS Gain Calibration from the displayed selections.
2. Perform Collimator Calibration by selecting Scanner Utilities on bottom of the left monitor then selecting Collimator Calibration from the displayed selections.
1. Perform Detailed Calibration by selecting Scanner Utilities on bottom of the left monitor then selecting Detailed Calibration from the displayed selections. Highlight the 8 Slice modes, customer preferred KVs, all SFOVs, small and large focal spot.
2. Perform Auto CT Number Adjust by selecting Scanner Utilities on bottom of the left monitor then selecting Auto CT Number Adjust from the displayed selections.
3. Verify 8 Slice operation by performing the tests detailed in Appendix A of this document.

Section 8 - System Checkout and Save State

1. Refer to the LFC Manual, and perform a Save System State. This will save the new calibrations as well as the customer and service protocols.
2. Refer to the LFC Manual, and perform the system performance checks as described in the manual. Ensure all networking and filming processes are still functional.
3. Complete and return Red Installation Product Locator Cards.

Appendix A

System Image Performance Verification

Image Acquisition:

In this section we will acquire two image series for evaluation in the next section.

Acquiring the 48cm Phantom Image Series

1. Mount the Phantom Holder on the head-end of the table.
2. Mount the 48cm Phantom on the Phantom Holder.
3. Align, level, & center the 48cm Phantom.
 - Align phantom using the internal laser lights.
 - Level phantom using bubble level and the Z Axis knob on the Phantom Holder.
 - Center phantom using the **Center Phantom** procedure in the left head **Scanner Utilities** selection and the X and Y Axis knobs on the Phantom Holder.
4. Set up the system to scan a single, 4 image, 48cm Phantom series.

Manual Scan Protocol Setup

Refer to Table 1 and set-up an Axial scan with the parameters shown.

Table 1 LightSpeed QX/I 48cm Phantom Image Series Scan Parameters

Series Description	Scan Type	Start Loc.	End Loc.	Total # of Images	Thick Speed	Interval (mm)	Gantry Tilt	SFOV	kV	mA	Total Exposure Time	DFOV (cm)	Recon Type
48 Lg Series 1	Axial Full 1.0 sec.	I7.50	S7.50	8	2.5 8i	0.00	S0.0	Large	120	340	1.0 sec.	50.0	Std

Auto Scan Protocol Setup

- a. On the Exam Rx desktop, select **New Patient**.
 - b. Type the following entries in the two listed **Patient Information** fields:
 - **Patient ID:** Service
 - **Name:** 48cm Phantom
 - c. From the Protocol display, click on the infant box.
 - d. On the Infant display window, click on the area below the infant's left foot to display the **miscellaneous** menu.
 - e. Click on the **20:10 Image Series 48 cm** button.
 - f. Use the first series.
5. Set internal Landmark.
 6. Acquire the default 32 image series of the 48 cm Phantom.

Acquiring the 20cm QA Phantom Image Series

1. Mount the Phantom Holder on the head-end of the table.
2. Mount the 20cm QA Phantom on the Phantom Holder.
3. Align, level, & center the 20cm QA Phantom.
 - Align black line on phantom using the internal laser lights.
 - Level phantom using bubble level and the Z Axis knob on the Phantom Holder.
 - Center phantom using the **Center Phantom** procedure in the left head **Scanner Utilities** selection and the X and Y Axis knobs on the Phantom Holder.
4. Set up the system to scan the QA#3 position on the 20cm QA Phantom image series.

Manual Scan Protocol Setup

Refer to Table 2 and set-up an Axial scan with the parameters shown.

Table 2 LightSpeed QX/I 20cm QA Phantom Image Series Scan Parameters

Series Description	Scan Type	Start Loc.	End Loc.	Total # of Images	Thick Speed	Interval (mm)	Gantry Tilt	SFOV	kV	mA	Total Exposure Time	DFOV (cm)	Recon Type
QA#3	Axial Full 2.0 sec.	S58.10	S66.85	8	1.25 8i	0.00	S0.0	Small	120	200	2.0 sec.	25.0	Std

Auto Scan Protocol Setup

- a. On the Exam Rx desktop, select **New Patient**.
 - b. Type the following entries in the listed **Patient Information** following fields:
 - **Patient ID:** Service
 - **Name:** 20cm QA#3 Phantom Image Series
 - c. From the Protocol display, click on the infant box.
 - d. On the Infant display window, click on the area below the infant's left foot to display the **miscellaneous** menu.
 - e. Click on the **20:12 Image Series QA** button.
 - f. When the scan prescription appears on the left monitor, skip to the QA#3 / small / 8X1.25 series (displayed in the series description box) by pressing the next series button.
5. Set internal Landmark.
 6. Acquire the QA#3 position of the 20cm QA Phantom image series.

Image Evaluation:

In this section, we will utilize the Auto Image Analysis tool to evaluate the 48 cm and QA images series'. This requires Class C software to be loaded on the scanner and the service key to be installed.

Image Analysis of the 48cm Phantom Image Series

1. Open the Service Desktop.

Note: <ALT> F1 will bring the window that the cursor is over to the front of the display area.
<ALT> F2 will push the window that the cursor is over to the back of the display area

2. From the service image browser, select the **48cm / 8X2.5 / 1s** image series from the 48cm Phantom exam acquired in the previous section.
3. Select the Image analysis process for the 48cm series by performing the following steps.
 - a. Select the Image Quality tab on the Service Desktop.
 - b. Select Image Analysis from the list provided user the Image Quality tab.
 - c. On the Image Analysis GUI, select the following pulldown items.

Auto 1X

48cm Phantom

48cm / 8X2.5 / 1s

auto

- d. On the Image Analysis GUI, select the **Accept** button.
- e. The system will process the image data and produce a report indicating a PASS / FAIL for each test. All tests should indicate PASS. A FAIL indication requires further investigation or calibration.

Image Analysis of the QA Phantom Image Series

1. Open the Service Desktop.

Note: <ALT> F1 will bring the window that the cursor is over to the front of the display area.
<ALT> F2 will push the window that the cursor is over to the back of the display area

2. From the service image browser select the **20:12 Image** image series from the 20cm QA#3 Phantom exam acquired in the previous section.
3. Select the Image analysis process for the 48cm series by performing the following steps.
 - a. Select the Image Quality tab on the Service Desktop.
 - b. Select Image Analysis from the list provided user the Image Quality tab.
 - c. On the Image Analysis GUI, select the following pulldown items.

Auto 1X

QA Phantom

QA#3 / 8X1.25 / 2s

auto

- d. On the Image Analysis GUI, select the **Accept** button.
- e. The system will process the image data and produce a report indicating a PASS / FAIL for each test. All tests should indicate PASS. A FAIL indication requires further investigation or calibration.